

Exponentially Biased Debt Repayment Weight

A Lightweight, Interpretable Repayment Ranking Heuristic

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***Status.** Independent technical note. This document has not been peer-reviewed. The method is intended as a transparent repayment-ranking heuristic and should not be treated as a calibrated probability-of-default model, lending decision system, or legal, financial, or regulatory advice.*

Abstract

This technical note defines the Exponentially Biased Debt Repayment Weight, a closed-form method for ranking indebted users from repayment history. Each repayment is normalized by a fixed reference debt amount and weighted by an exponential recency function, producing a transparent score that emphasizes recent repayment behavior while retaining a controlled contribution from older history. The standard form uses an asymptotic exponential decay function. An extended form introduces boundary-condition coefficients that force the decay function to reach exactly zero at a specified cutoff period. The method requires no model training, labelled outcomes, demographic features, or third-party behavioural data. It is presented as an independent, non-peer-reviewed technical note and should be understood as a repayment-ranking heuristic rather than a calibrated credit-risk model.

Keywords: repayment ranking; credit scoring; collections prioritisation; heuristic; microfinance; interpretability; exponential decay; EWMA; recency weighting;

1 Introduction

The assessment of repayment behaviour is a recurring problem in lending, collections, and portfolio management. A borrower who repaid recently usually carries a different operational signal from a borrower who made the same repayment many periods ago. The purpose of this note is to define a simple mathematical method for representing that recency-sensitive signal.

The proposed method is not presented as a replacement for validated credit scoring systems or machine-learning models. Rather, it is a transparent ranking heuristic for settings in which data scarcity, infrastructure constraints, or explainability requirements make a closed-form method attractive. This positioning is consistent with the broader use of exponential smoothing and exponentially weighted moving averages in time-series analysis and financial risk management (Brown, 1956; Holt, 2004; J.P. Morgan/Reuters, 1996).

The contribution is the packaging of four established ideas into a practical repayment-ranking score: fixed repayment buckets, debt-normalized repayments, exponential recency weighting, and an optional hard cutoff. The method is simple enough to compute in a spreadsheet while retaining a precise mathematical definition.

2 Related Work

Exponential smoothing was introduced for forecasting by Brown (1956) and developed further by Holt (2004). Related exponentially weighted moving average methods are widely used in statistical process control and financial risk management (Montgomery, 2012; J.P. Morgan/Reuters, 1996). Behavioural credit scoring uses observed account behaviour, including repayment and delinquency patterns, to assess account risk (Thomas, 2009). In

modern practice, recency-sensitive features are often fed into statistical or machine-learning models rather than used directly as the score (Lessmann et al., 2015; Kennedy et al., 2013).

The distinction in this note is operational rather than purely mathematical. The exponentially weighted repayment sum is used directly as the ranking output. This makes the method easier to inspect and explain, but it also means that the score should not be interpreted as a probability of default, affordability estimate, or expected-loss model.

3 Method: Standard Form

The standard form assigns each repayment period an exponentially decaying recency weight. Let i denote a reverse-time period index counted backwards from the ranking date, so $i = 1$ denotes the most recent period, $i = 2$ denotes the immediately preceding period, and so on through $i = n$. A period may be a day, week, month, quarter, year, or any other fixed interval appropriate to the repayment schedule and available data. The period length should be defined consistently across all users in a ranking run. The plotted period origin corresponds to period index 1 and weight 1.0; if a user has no repayment in a period, that period should be recorded as $P_i = 0$ rather than omitted.

Table 1: Notation for the standard and extended repayment weighting methods.

Symbol	Definition
i	Reverse-time period index; $i = 1$ is the most recent period.
n	Number of historical periods included in the observation window.
k	Decay-rate constant controlling how quickly recency weights decline.
h	Half-life in periods; equivalent decay-rate choice given by $k = \ln(2)/h$.
L	Fixed reference debt amount used as the normalization denominator.
P_i	Repayment amount posted during period i .
d_i	Normalized repayment in period i .
b	Standard exponential bias weight.
b_i	Exponentially biased repayment contribution.
w	Composite debt repayment weight used for ranking.
C_0, C_1	Boundary-condition coefficients for the extended decay-to-zero form.

3.1 Normalised Repayment

Before applying recency weighting, repayment in each period is normalised by a fixed reference debt amount L . This amount serves as the normalization denominator in the repayment fraction. In implementation, L should be chosen consistently across users, for example as the outstanding balance at the start of the observation window or the total scheduled amount for the account.

$$d_i = \frac{P_i}{L}$$

The value d_i may be interpreted as the fraction of the chosen reference debt amount repaid during period i . A policy decision is required on whether overpayments may produce $d_i > 1$ or whether values should be capped.

3.2 Standard Exponential Decay

The standard decay function assigns the following bias weight to each historical period:

$$b = e^{-k(i-1)}$$

The offset $i - 1$ ensures that the most recent period receives weight one. As i increases, the weight declines asymptotically toward zero. The parameter k controls the decay rate: small k values preserve more historical memory, while large k values give stronger emphasis to recent behaviour.

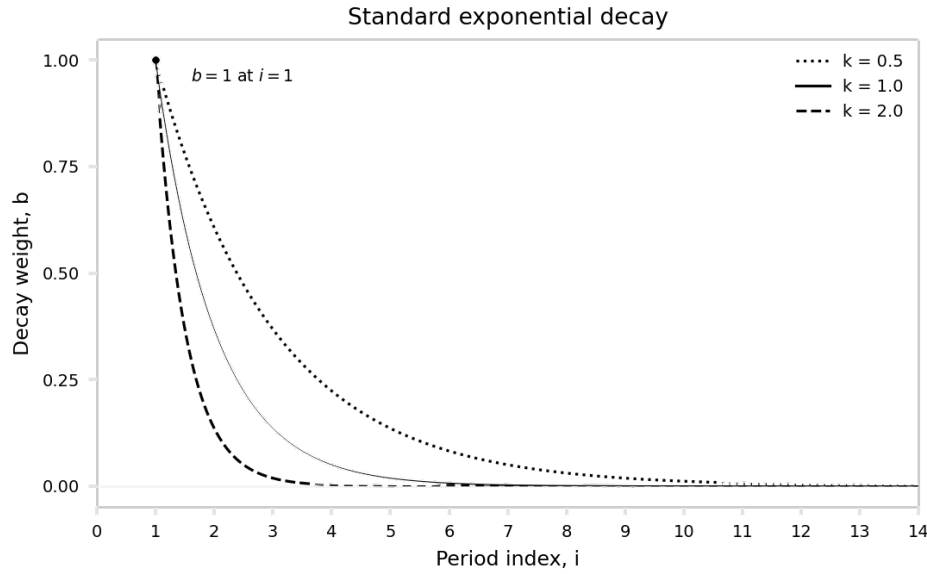


Figure 1: Standard exponential decay curves for $k = 0.5$, $k = 1.0$, and $k = 2.0$. Each curve equals 1.0 at period index $i = 1$ and approaches zero asymptotically for large i .

3.3 Exponentially Biased Repayment and Composite Score

The exponentially biased repayment contribution for period i is obtained by multiplying the normalised repayment by the standard decay weight:

$$b_i = d_i \cdot e^{-k(i-1)}$$

The composite score is the sum of the weighted repayment contributions across the observation window:

$$w = \sum_{i=1}^n b_i$$

Larger values of w indicate stronger recent repayment activity relative to the chosen reference debt amount and observation window. The score is dimensionless and should be interpreted as a ranking score, not as a probability.

3.4 Half-life Interpretation

The decay constant k can be selected directly, but a half-life is often easier to communicate. If h is the number of periods after which the recency weight should fall to one half, then:

$$k = \frac{\ln(2)}{h}$$

Table 2: Example half-life choices for the standard decay function.

Half-life h	k	Interpretation
1 period	0.693	Very strong recency bias
2 periods	0.347	Strong recency bias
4 periods	0.173	Moderate recency bias
8 periods	0.087	Gentle recency bias

4 Method: Extended Decay-to-Zero Form

The standard form approaches zero asymptotically but never reaches it. Some use cases require a strict historical cutoff, so that repayments at or beyond a specified period contribute exactly zero. The extended form preserves the same exponential shape while adding boundary-condition coefficients that force the curve to pass through $y(1) = 1$ and $y(n) = 0$.

Let n denote the cutoff period and k the decay factor. Define the coefficients C_0 and C_1 as follows:

$$C_0 = \frac{e^k}{e^k - e^{kn}}$$

$$C_1 = \frac{e^{kn}}{e^{kn} - e^k}$$

The active exponential segment is then:

$$y = C_0 + C_1 \cdot e^{-k(x-1)}$$

For repayment weighting, the extended method is a finite-window rule: the exponential segment is used only before the cutoff, and the weight is zero at the cutoff and thereafter.

$$y(x) = C_0 + C_1 \cdot e^{-k(x-1)}, \quad 1 \leq x < n$$

$$y(x) = 0, \quad x \geq n$$

The graph should therefore be read as a truncated decay-to-zero weighting function. The analytic expression is not continued beyond the cutoff period n , because that continuation would fall below zero and no longer represent a repayment weight.

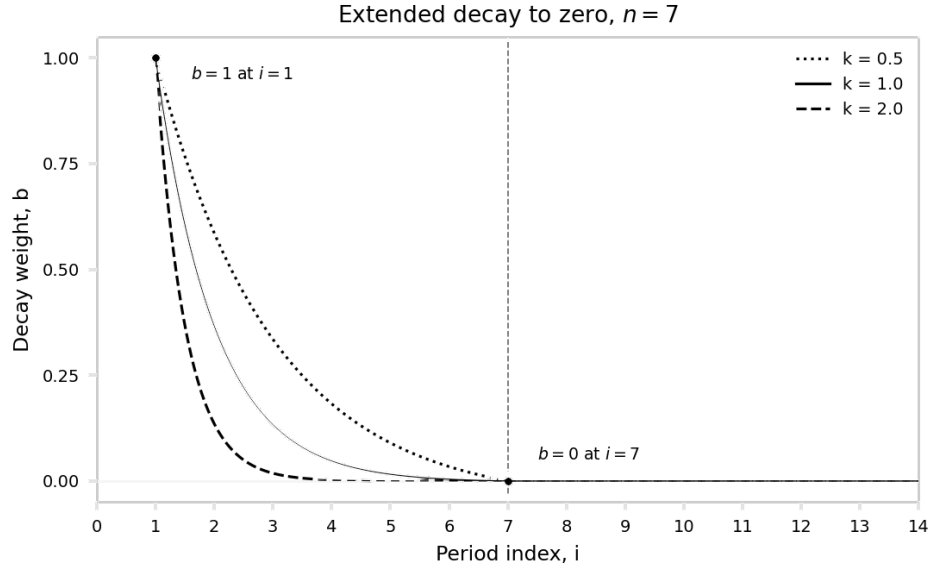


Figure 2: Extended decay-to-zero curves for $k = 0.5$, $k = 1.0$, and $k = 2.0$, with cutoff $n = 7$. Each curve equals 1.0 at $x = 1$ and reaches exactly 0.0 at $x = n = 7$.

The full expression for the active exponential segment, obtained by substituting the coefficients into the extended form, is:

$$y = \frac{e^k}{e^k - e^{kn}} + \frac{e^{kn}}{e^{kn} - e^k} \cdot e^{-k(x-1)}$$

This extended form is useful when historical repayments outside the chosen window should have no influence at all. It should be selected intentionally, because a hard cutoff changes the interpretation of older history relative to the standard form.

5 Worked Example

Consider a six-period observation window with $L = 1000$ and $k = 0.35$. For concreteness, take each period to be one week; period $i = 1$ is the most recent period. The repayment sequence is $P = [120, 80, 0, 50, 0, 100]$.

Table 3: Worked example for the standard repayment weighting method.

i	P_i	d_i	Decay weight	Contribution b_i
1	120	0.120	1.000	0.120
2	80	0.080	0.705	0.056
3	0	0.000	0.497	0.000
4	50	0.050	0.350	0.017
5	0	0.000	0.247	0.000
6	100	0.100	0.174	0.017

The resulting standard-form score is $w = 0.211$. The normalised variant $w_{\text{norm}} = w / \sum_i b$ is 0.071; this variant may be useful when comparisons across different observation-window lengths are unavoidable.

Table 4: Effect of repayment timing when total repayment is unchanged.

User	Repayments $P_1 \dots P_6$	Total	w	Interpretation
A	180, 120, 0, 0, 0, 50	350	0.273	Repayments concentrated in recent buckets.
B	50, 0, 0, 0, 120, 180	350	0.111	Same total repayment, but most repayment is older.

6 Discussion

The standard and extended forms serve related but distinct purposes. The standard form is appropriate when older repayments should continue to have a diminishing influence. The extended form is appropriate when a hard historical boundary is required. In both cases, the score remains transparent: each contribution can be traced to a repayment amount, a reference debt amount, and a recency weight.

The method is computationally light. Scoring N users across n periods requires $O(Nn)$ operations and no model training. This makes the method suitable for spreadsheets, database queries, and simple scripts. Its simplicity is also its limitation: the score does not estimate probability of default, affordability, expected loss, hardship, fraud, disputes, or vulnerable-customer status.

The score should therefore be used as a prioritisation signal rather than as a sole decision rule. In consequential settings it should be paired with governance, human review, compliance checks, and empirical validation.

7 Implementation Notes

Implementation quality matters because inconsistent bucket construction or reference debt amount selection can dominate the apparent precision of the formula. Table 5 summarises recommended handling for common edge cases.

Table 5: Practical implementation cases.

Case	Recommended handling
Missed repayment period	Set $P_i = 0$; do not omit the period.
No repayment history	The score is zero when every P_i is zero.
$L \leq 0$	Undefined; exclude or resolve the account before scoring.
Negative repayment or refund	Treat as an accounting adjustment or exclude by policy.
Overpayment	Decide whether d_i may exceed one or whether values should be capped.
Irregular repayment dates	Bucket payments into fixed intervals before scoring.
Different history lengths	Use the same observation window across every user in a ranking run.

8 Conclusion

This technical note has defined the Exponentially Biased Debt Repayment Weight, a closed-form repayment-ranking heuristic based on normalised repayments and exponential recency weighting. The standard form provides a simple

asymptotic decay model. The extended decay-to-zero form preserves the same basic intuition while forcing the weight to reach exactly zero at a chosen cutoff period.

The value of the method lies in transparency and ease of implementation rather than predictive sophistication. Because it is not a calibrated probability-of-default model, it should be validated carefully before use in any consequential lending or collections process.

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